RM9: Approved section-speed observer
Current-task history audit and its limits

Method-first ramp-metering review

5 October 2026

Disposition. Observer C is approved; UG4-O is closed. This bounded R2 pass establishes conditional information sufficiency, not a positive equal-history/useful-action certificate or universal impossibility. R3 admits no rescue assumption. R5 screens alternatives only; UG5 requires a user disposition before a changed task is formulated. No traffic experiment is run.

1 Approved observation, unchanged traffic law

For a mainline cell of length L, write lane masses n_1, n_2 in vehicle equivalents, total $N = n_1 + n_2$, density $k_\ell = n_\ell/L$ and storage M_ℓ. Public FD parameters are v_ℓ, w_ℓ, C_ℓ and $k_\ell^{\mathrm{jam}} = M_\ell/L$. Define $c_\ell = v_\ell/L$, $\gamma_\ell = w_\ell/L$ and the constitutive equilibrium rate

$$g_\ell(n) = \min\{c_\ell n, C_\ell, \gamma_\ell(M_\ell - n)\}. \tag{1}$$

With $U_\ell^{\mathrm{eq}}(k) = f_\ell^{\mathrm{eq}}(k)/k$ for occupied lanes, C observes

$$V = \frac{n_1 U_1^{\mathrm{eq}}(k_1) + n_2 U_2^{\mathrm{eq}}(k_2)}{N}, \qquad \frac{NV}{L} = g_1(n_1) + g_2(n_2), \quad N > 0. \tag{2}$$

Empty-lane contributions are zero without evaluating 0/0; an empty cell's speed is unavailable. This is a homogeneous representative-equilibrium closure, not a detector or receiving-limited physical-speed theorem. The right side is *not* the actual accepted face rate, which is observed separately.

Preserve the adopted ordering: bounded lateral transfer conserving mass, then recompute forward sending and receiving, then conservative forward/merge updates. The Default positive-priority allocator redistributes unused receiving; the mainline and ramp share their forward receiver. Meter permission, actual release and actual downstream merge entry remain different where storage lies between them. No new traffic state or law follows from the observer approval.

TST qualification. C describes traffic present in a section. It does not compute time spent in the whole system. For time step h and all-stream present population P^s,

$$J_h = h \sum_{s=0}^{S-1} P^s. \tag{3}$$

P^s includes mainline, pre/post-meter ramp storage and external waiting, with the same arrivals and clearance rule for every method. Normalization by a fixed common total mass does not change action ranking. Do not replace this objective by a cell length divided by V; the FD closure need not describe actual blocked motion. The same-region time–space speed identity is a different statistic.

2 Branch-local information and global ambiguity

With $z = n_1$ and observed N, feasible splits lie in

$$I_N = [\max(0, N - M_2), \min(M_1, N)], \qquad G_N(z) = g_1(z) + g_2(N - z) = NV/L. \tag{4}$$

Use F for the free branch, P for a capacity plateau and H for the congested branch. Away from branch boundaries, the lane-rate slopes are respectively $c_\ell, 0, -\gamma_\ell$. The slope of G_N is the difference of lane slopes:

Lane 1 / lane 2	F	P	H
F	$c_1 - c_2$	c_1	$c_1 + \gamma_2$
P	$-c_2$	0	γ_2
H	$-(\gamma_1 + c_2)$	$-\gamma_1$	$\gamma_2 - \gamma_1$

A nonzero slope determines at most one candidate *inside that known branch domain.* Equal free-flow mobilities or equal congested wave coefficients make the respective same-branch measurement constant; plateau/plateau is also constant when that branch actually exists. These are algebraic conditions, not permission to impose equal parameters to manufacture a collision.

Global warning. Each g_ℓ is concave; G_N is concave and piecewise affine. Roots can occur on opposite sides of its maximum, even when each involved branch has a nonzero slope. Flat intervals can give a continuum. Thus a free-flow inversion is a global identification result only when all other feasible branches are excluded by public state/domain information or completed observations. We do not silently observe branch membership. Nonconcave changes would require a new law, not use of this argument.

Plateau warning. A positive-width P branch requires

$$C_\ell < \frac{c_\ell \gamma_\ell M_\ell}{c_\ell + \gamma_\ell}. \tag{5}$$

For the ordinary triangular FD, equality holds and P is only a peak point. FB0030's triangular capacity relation does not justify a positive interval with both lanes on P. A separately capped trapezoid could be a legitimate public model if independently supported; it has not been adopted here.

Actual faces. Even a compatible section-speed root must reproduce completed actual aggregate face counts/rates, ramp balances, all other cells and prior controls under the adopted lateral/forward law. A capped face alone does not remove the separate section-speed information. Future action-induced flow cannot be used to decide which root existed before choosing that action.

3 The complete-history requirement

Let x^s denote the complete modeled mainline/ramp state, p public parameters, b^s all declared forcing/boundary allocation, and u^s prior meter permissions. The deterministic adopted update is

$$x^{s+1} = F_B(x^s, b^s, u^s; p). \tag{6}$$

Let O_C include stored-state inventories/section speeds and completed labeled face and ramp observations. At the decision epoch, define the feasible initial state set using the *whole common declared history* H_*:

$$\mathcal{X}(H_*) = \{x^0 : \; x^{s+1} = F_B(x^s, b^s, u^s; p), \; O_C(x^s, \text{completed fluxes}) = O^s_{\text{admitted}} \text{ for all } s \le *\}. \tag{7}$$

All candidates obey storage, geometry, conservation and public forcing. This defines the proof obligation; it is not a numerical estimator or a solved site-specific constraint system.

Conditional sufficiency lemma. If the admitted information uniquely identifies the complete state at any common past epoch $a \le *$, then all later states up to the decision are uniquely determined. Proof: apply the public deterministic update once to the known state and inputs; repeat by induction. For any common future schedule and forcing, the same induction gives identical future states, exit histories and TST, hence identical action rankings.

This includes a fully known complete initialization (including a genuinely known empty system), or an earlier epoch where every state coordinate is uniquely reconstructed. Queue formation, capacity caps and later snapshot ambiguity do not invalidate this result. A uniquely invertible local branch in one cell does not by itself identify the whole system.

Unresolved domain. If only aggregate initial information is known and multiple complete states survive *all* declared observations, the lemma does not settle the task. Mixed/congested branches may permit snapshot ambiguity, but no genuine site/initialization-supported pair has been established here that survives the coupled full history and changes useful common actions. Conversely, no global impossibility theorem has been proved for every such domain. Do not make the history short or forcing private just to obtain multiplicity.

If a noninjective lateral update maps two different pre-states to the same *complete* state before actions act, with all remaining inputs equal, their lost coordinate cannot create a subsequent action distinction. A failure of a memoryless total-density closure is also not sufficient: a stronger aggregate-history method can reconstruct an otherwise missing coordinate.

4 Action and gradient status; next physical comparison

The required positive claim would need two genuine feasible states with identical H_*, distinguishable useful actions from the same admissible open-loop family, and complete affected-population time accounting. None is admitted by this pass. Therefore R4 has no new certificate to propagate into a claim of useful gradients. Existing branch-wise conceptual derivatives are conditional; no finite-difference, cross-fidelity or one-step usefulness test has been conducted in this methodology review.

R3 declines unsupported symmetry, invented plateau widths, weakened/noisy speed observations, omitted waiting, private forcing and extra lanes introduced only to defeat an inverse. These could be legitimate features of another source-grounded task, but need explicit scientific admission and cannot be silent repairs of this one.

R5's bounded source comparison supports physical questions, not positivity:

- Laval, Cassidy and Daganzo's inspected merge model represents bounded acceleration and associated voids, even at a site without an acceleration lane. Its ramp-queue-dependent entry speed is not automatically hidden when that queue is known. No exact RM9 history collision follows.
- Srivastava's 2016 dissertation, Chapter 3.3.1, changes congested demand and adds a jam-demand parameter while retaining supply and density conservation. This is an aggregate-law competitor, in a queue-discharge task; the separate lane-drop model in Chapter 4 requires additional merge modeling for a ramp extension. Neither is copied into RM9.
- A real finite acceleration lane may require position/urgency and changed geometry. Gap-informed metering changes sensing and temporal action semantics. Both require their own task comparison, not an unchanged-law counterexample.

Recommendation at UG5. Accept the scoped absence of an established positive result for this bounded current-task pass, and discuss a targeted recovery/acceleration-family review beginning with the strongest credible aggregate demand/history competitor. Add state only if necessary to represent independently supported information that this competitor cannot reconstruct. This is a proposed next scope, not an adopted law, successor or universal rejection of the current family. The companion findings present fair alternatives.

Primary provenance and reviewer boundary

FD source: FB0030, final primary lane-model PDF, capacity relation printed 1768. Physical merge mechanism: Laval, Cassidy and Daganzo, inspected merge paper, Sections 2/4. Aggregate competitor: Srivastava, UCI dissertation 2016, selected Chapters 3/4. Observer provenance remains in the R1 source matrix. Equations and history lemmas above are reviewer deductions under the explicitly approved closure, not literature theorems for the combined adapted task. Search/access and claim ledgers retain version and coverage limits. No source establishes the combined hidden-information, useful-action and useful-gradient chain.